Predictive Modeling of Housing Prices in the Philippines Using Simple and Multiple Linear Regression

Luke Joaquin Bernardo
College of Computer Studies and Engineering
Jose Rizal University
Taguig City, Philippines
lukejoaquin.bernardo@my.jru.edu

***Abstract*—This paper presents the research on the use of Simple and Multiple Linear Regression in order to make house price predictions in the Philippines, based on the dataset containing 2,999 real estate properties. My objective was to ascertain whether the complexification of the model, i.e., adding characteristics: the number of bedrooms, the number of bathrooms, the size of the land, etc. can lead to a higher predictive power than the simple model which takes into consideration only the floor area. Having performed stringent data cleaning to address missing values and noise, the experimental results were not expected to achieve the performance as observed: the Simple Linear Regression model ($R^2 = 0.3459$), was marginally better than the Multiple Linear Regression model ($R^2 = 0.3372$). This is an indication that floor area is the main predictor of price in this dataset and inclusion of correlated variables did not add significant prediction of price to the model, but created multicollinearity.**

***Keywords*—Linear Regression, Predictive Modeling, Housing Prices, Data Science, Machine Learning**

I. Introduction

Price forecasting is a crucial instrument to both buyers and sellers in the fast changing real estate market as well as the investors. Predictive modeling enables an individual to do away with the aspect of guesswork and predict property value using historical information. This paper shows how the end-to-end data science pipeline can be implemented in practice, i.e. how Linear Regression methods can be used to clean up raw data and evaluate the final model.

Linear Regression is among the basic statistical tools used to model the relationship between an objective variable (Price) and predictors (independent variables). I contrast two methods: Simple Linear Regression (SLR), where only one strong predictor is used, and Multiple Linear Regression (MLR), where several variables are taken into consideration. The major objective of the paper is to explore the necessity of a more complex model to produce a more accurate result or a simplistic approach can be used in this particular dataset.

II. Data Collection

The research data in this research is the Housing Prices Dataset Philippines, which was obtained in the Kaggle free database.

A. Dataset Description

The original raw data had 2,999 records and 12 fields. It contained some of the important attributes of the property i.e. Price (the target variable), Floor Area, Land Size, Bedrooms, Bathrooms and Car Spaces.

B. Data Cleaning and Preprocessing

Data acquired in the real world is not often prepared to be analyzed instantly. Primary examination showed that there was severe noise, which was to be removed by using Python (Pandas), and the following cleaning steps were made:

1) **Target Validation:** I eliminated 75 rows with null Price value. These rows were not useful since supervised learning entails that a learner is provided with a labeled target to learn.
2) **Feature Pruning:** I removed columns that had too much missing data, namely `Rooms (total)` (83% missing) and `Build (Year)` (82% missing) because they only had insufficient data to use.
3) **Imputation:** I imputed all missing values instead of dropping them to retain as much of the data as possible. Absence of `Car Spaces` was taken to be 0. In the case of `Bedrooms`, `Bathrooms` and `Floor Area`, I used median as the filler of any gaps created which is resistant to outliers.
4) **Last Polish:** Text based columns were coded or dropped leaving behind a clean dataset of 2,924 rows to model.

III. Attribute Selection

Attribute selection is the process of choosing the most relevant variables to improve model performance and reduce overfitting.

A. Correlation Analysis

A correlation matrix was generated to identify the variables most strongly related to `Price`. The analysis revealed the following Pearson correlation coefficients:

- **Floor Area:** 0.58 (Strongest Positive Correlation)
- **Bathrooms:** 0.41
- **Bedrooms:** 0.38
- **Land Size:** 0.35

Based on these results, `Floor Area` was selected as the sole predictor for the Simple Linear Regression model. For the Multiple Linear Regression model, five features were selected: `Floor Area`, `Bedrooms`, `Bathrooms`, `Land Size`, and `Car Spaces`.

IV. Model Building

In order to achieve a fair assessment, I divided the cleaned dataset in Training Set (80%), and Testing Set (20%) with a random seed of 42. This guarantees the reproducibility of the results.

A. Simple Linear Regression (SLR)

The SLR model was modeled on the basis of Floor Area only. The model is trying to determine the optimum straight line ($Y = mX + b$) explaining the rise in price with increased floor area.

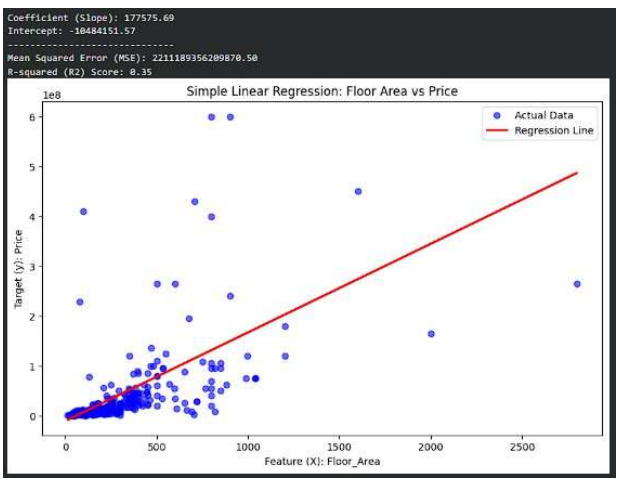

Fig. 1. Simple Linear Regression Fit: Floor Area vs. Price.

B. Multiple Linear Regression (MLR)

All the five selected features were used to train the MLR model. This model tries to predict Price by a multi-dimensional hyperplane that is generated by the joint effect of all variables instead of a line.

V. Discussion

I compared both models on the unknown Testing Set on the basis of common measures: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2).

A. Performance Comparison of Models

The results are summarized in Table I.

TABLE I
Model Performance

Metric	Simple Regression	Multiple Regression
R^2 Score	**0.3459**	0.3372
RMSE (PHP)	**47.02 Million**	47.33 Million
MAE (PHP)	17.87 Million	**17.64 Million**

Remarkably, Simple Regression model had a little better score (0.346) in comparison to Multiple Regression model (0.337). This makes a hole in the usual notion that more data is better. Adding additional variables in this case, however, did not enhance the model to explain the variance in the housing prices.

B. Coefficient Analysis: The Negative Bathroom Anomaly

A checkup of the coefficients of the MLR showed that although the coefficients of the Floor Area and Bedrooms were positive, those of Bathrooms showed an important negative coefficient (-5.46 million).

In an intuitive sense, the value of a home should be boosted when one adds a bathroom. This is a characteristic of multicollinearity. Big houses are usually characterized by big floor plan and numerous bathrooms. There is a high likelihood that the model did not manage to isolate the contribution of having a bathroom compared to the contribution of having more space, which resulted into skewed mathematical weights.

C. Residual Analysis

I had to plot the residuals (difference between the actual price and the predicted price) to verify the number of errors.

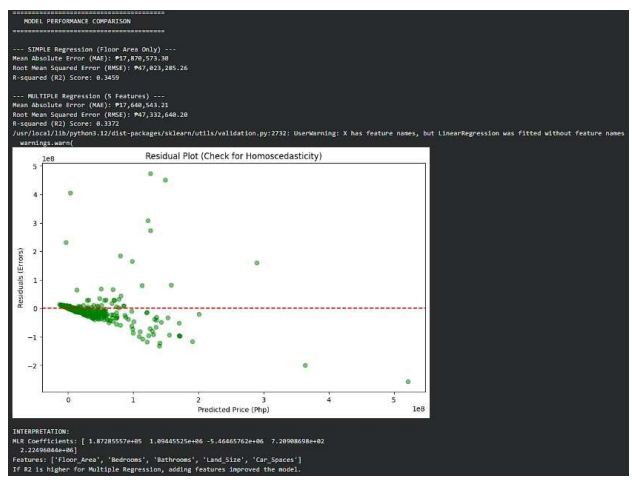

Fig. 2. Residual Plot showing Heteroscedasticity (Funnel Shape).

There was a characteristic funnel shape of the plot in that the number of errors was minimal with cheaper houses and huge with expensive ones. This proves heteroscedasticity. It indicates that a linear model can be effective in averagely priced houses, but not among upscale luxury houses, which can take a nonlinear trend.

VI. Conclusion

This paper demonstrates one major insight in data science, which is that model complexity is not necessarily the guarantee of improved performance. Using simple Linear Regression with the only Floor Area the complex Multiple Linear Regression was a little less effective. This must have been a result of noise and multicollinearity that features such as bathrooms and bedrooms induced to the model instead of assisting the model.

These findings could be improved in the future by using a Log Transformation on the Target variable to correct the heteroscedasticity in the residual plot, or regularization methods such as Ridge Regression to better deal with the correlated features.